Unit: Unit 3: Functions **Name:** ____________

Topic: Definition of Relation and Function, Domain, Range, and Notations **Date:** __________

Foundation (Jalapeno)

Q1. Which of the following is a relation?
(A) A set of ordered pairs
(B) A set of integers
(C) A graph of a line
(D) A table of values
(E) All of the above

Q2. What is the vertical line test used for?
(A) To determine the domain of a function
(B) To determine the range of a function
(C) To determine if a relation is a function
(D) To graph a function
(E) To solve an equation

Q3. Given the function $f(x) = x^2$, what is the domain?
(A) All real numbers
(B) All integers
(C) All positive numbers
(D) All negative numbers
(E) None of the above

Q4. What is the range of the function $f(x) = |x|$?
(A) All real numbers
(B) All positive numbers
(C) All negative numbers
(D) All integers
(E) None of the above

Q5. Given the equation $y = 2x + 1$, what is $f(2)$?
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7

Q6. Which of the following is a function?
(A) A relation with one output for each input
(B) A relation with one input for each output
(C) A relation with multiple outputs for each input
(D) A relation with multiple inputs for each output
(E) None of the above

Q7. Given the graph of a function, how can you find the domain?
(A) Look at the x-values of the points on the graph
(B) Look at the y-values of the points on the graph
(C) Look at the x-intercepts of the graph
(D) Look at the y-intercepts of the graph
(E) None of the above

Q8. What does the notation $f(x)$ represent?
(A) The input of a function
(B) The output of a function
(C) The graph of a function
(D) The equation of a function
(E) The domain of a function

Open Ended Questions (FRQ)

Q9. Given the function $f(x) = x^2 - 4$, find the domain and range.

Q10. Use the vertical line test to determine if the relation $y = |x|$ is a function. Explain your reasoning.

Chef's Tips

- Tip 1: Make sure to read each question carefully.
- Tip 2: Use the vertical line test to determine if a relation is a function.
- Tip 3: Check your work and explain your reasoning.